

Node RED

Practicals

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1-Node-RED Lab Exercise: Processing IoT Sensor Data

Objective

In this lab, you will process an IoT dataset that has already been fetched. You will use the following Node-RED nodes:

- **JSON (Parse)**
- **Split**
- **Change**
- **Range**
- **Join**
- **Debug**

Exercise 1: Parse the IoT Dataset

Objective

Convert the incoming JSON string into a JavaScript object.

Flow

Inject → JSON → Debug

Tasks

- Import the provided IoT dataset.
- Use the **JSON** node to parse the data.
- Display the parsed object in the **Debug** panel.

Expected Result

The Debug panel should display an array containing **5 IoT device objects**.

Exercise 2: Split the Dataset

Objective

Process one IoT device at a time.

Flow

Inject → JSON → Split → Debug

Tasks

- Use the **Split** node to separate the array.
- Verify that each output message contains information for only one device.

Expected Result

The Debug panel should display **five individual messages**, one for each device.

Exercise 3: Modify Data Using the Change Node

Objective

Create and modify message properties.

Flow

Inject → JSON → Split → Change → Debug

Tasks

Using only the **Change** node:

- Create a property named **msg.device** and copy the value from **DeviceID**.
- Create a property named **msg.location** and copy the value from **Location**.
- Remove the **Timestamp** field from the payload.

Expected Result

Each message should contain the new properties while the timestamp is removed.

Exercise 4: Convert Temperature Using the Range Node

Objective

Scale temperature values using the **Range** node.

Flow

Inject → JSON → Split → Range → Debug

Tasks

Convert the temperature values from the range:

- 0°C → 100°C

to

- 32°F → 212°F

Expected Result

Example:

Celsius Fahrenheit

24.5°C ≈ 76.1°F

35.7°C ≈ 96.3°F

Exercise 5: Display Only Critical Devices

Objective

Filter the dataset.

Flow

Inject → JSON → Split → Switch → Debug

Tasks

Display only devices whose **Status** is **Critical**.

Expected Result

Only the Factory Floor device (**DEV005**) should appear in the Debug panel.

Exercise 6: Join the Messages

Objective

Rebuild the original dataset.

Flow

Inject → JSON → Split → Change → Join → Debug

Tasks

After modifying each device message, use the **Join** node to combine them back into a single array.

Expected Result

The Debug panel should display one array containing all processed devices.

Exercise 7: Create Battery Status

Objective

Generate a new property using the **Change** node.

Flow

Inject → JSON → Split → Change → Debug

Tasks

Create a new property named **BatteryStatus** according to the following rules:

Battery Level Battery Status

Above 80% Excellent

60%–80% Good

40%–59% Low

Below 40% Replace Soon

Expected Result

Device Battery Status

DEV001 92% Excellent

DEV002 80% Good

DEV003 67% Good

DEV004 55% Low

DEV005 30% Replace Soon

Exercise 8: Simplify the Device Information

Objective

Create a cleaner message format.

Flow

Inject → JSON → Split → Change → Debug

Tasks

Transform each device into the following structure:

```
{  
  "device": "DEV001",  
  "location": "Warehouse A",  
  "temperature": 24.5,  
  "humidity": 58,  
  "status": "Active"  
}
```

Use only the **Change** node.

Exercise 9: Count the Devices

Objective

Use **Split** and **Join** together.

Flow

Inject → JSON → Split → Join → Debug

Tasks

Split the dataset into individual messages and then join them back into one array. Determine the total number of devices in the final array.

Expected Result

The total number of devices should be 5.

Exercise 10: Prepare Data for a Dashboard

Objective

Create a display-friendly object.

Flow

Inject → JSON → Split → Change → Debug

Tasks

Convert each device into the following format:

```
{
  "Device": "DEV001",
  "Temperature": "24.5 °C",
  "Humidity": "58 %",
  "Battery": "92 %",
  "Status": "Active"
}
```

Challenge Exercise: Complete IoT Data Processing Flow

Objective

Build a complete processing pipeline using all the nodes covered in this lab.

Flow

Inject → JSON → Split → Change → Range → Switch → Join → Debug

Requirements

- Parse the IoT dataset using the **JSON** node.
- Split the dataset into individual device messages.
- Convert the temperature values from Celsius to Fahrenheit using the **Range** node.
- Rename **DeviceID** to **Device**.
- Remove the **Timestamp** field.
- Keep only devices whose **Status** is **Active**.
- Join the filtered messages back into a single array.
- Display the final processed dataset in the **Debug** panel.

Expected Learning Outcomes

After completing this lab, you should be able to:

- Parse JSON data.
- Split arrays into individual messages.
- Modify message properties using the Change node.
- Scale numeric values using the Range node.
- Filter messages based on conditions.
- Join multiple messages into a single array.
- Verify outputs using the Debug node.
- Build a complete Node-RED data-processing workflow for IoT applications.

